

GROUNDHOG

e-Book

Digital Mining and the Path to Green Mining



How Data, Discipline, and Execution Unlock
Sustainable, High-Performance Operations

PUBLISHED BY:
GroundHog

groundhogApps.com

About the Author

Amitava is a seasoned Mining Engineer with a demonstrated history of working in Mining Operations, Mine Planning, Technical Services, Mining Consultancy, and Mine Digitization Globally. As Director of Product Operations at GroundHog, he drives Product Operations, Digital Transformation, and Customer Success and supports Product Development and Business Development Strategy.



Mr. Dutta has deep expertise in the implementation of Mine Digitization, Mine Automation, and Process Efficiency Improvement for highly mechanized mines. He has demonstrated capabilities in Mine Digitization, Mine Planning Software, Fleet Management Solutions, Mining Simulation, Equipment Asset Management, Asset optimization, and many Business Excellence Tools. Successfully implemented many cost-saving and energy-saving projects for mining process efficiency improvement.



Executive Summary

Digital mining and green mining are converging to create a new paradigm in the industry. As demand for critical minerals surges, mines must boost output while drastically cutting environmental impact. This eBook explains how data-driven operations, disciplined processes, and rigorous execution make this possible. It shows how sensors, analytics, and automation transform traditional mining into a sustainable, high-performance operation.

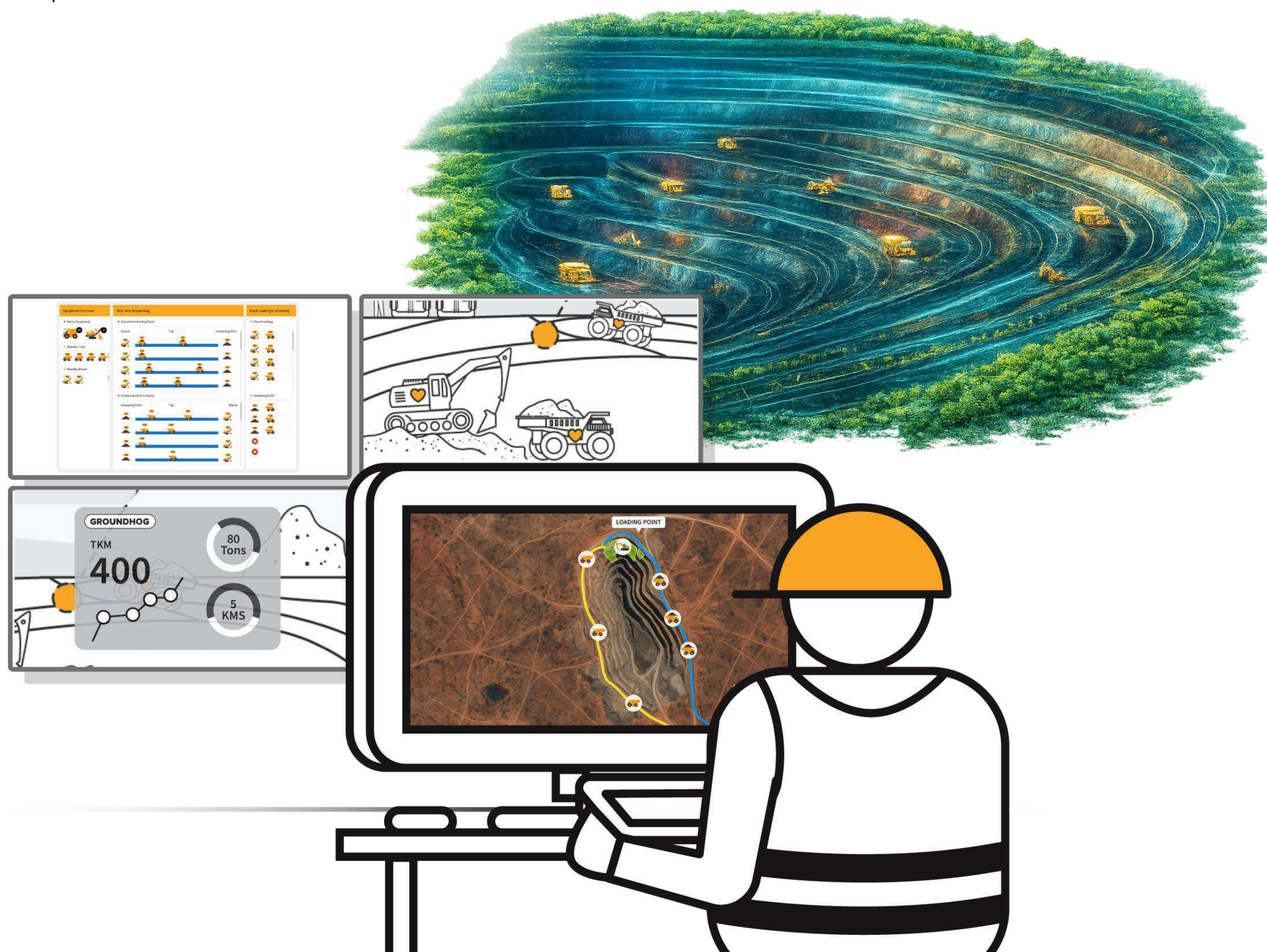
In modern mining, efficiency and sustainability go hand in hand. Reliable data captures every aspect of operations, from fleet movements to ore grades. When data is accurate and contextualized, teams make better decisions, reducing waste and saving energy. We explore the technologies and practices that turn raw data into actionable information, targeting every area of a mine: drilling, hauling, maintenance, environmental controls, and worker safety.

Each chapter drills into a specific topic. We start by explaining why green mining is now a business imperative. Then we define “digital mining” and emphasize the foundational role of data quality. We show how optimizing fleet movements and controlling fuel use becomes a prime “fuel-intensity” lever. We describe precision drilling and grade control that minimize waste. We cover execution systems like short interval control that lock in operational discipline. We highlight predictive maintenance and its impact on equipment life. We delve into water and tailings management with digital controls, sensors for environmental monitoring, and automation’s role in safer outcomes.

Throughout, we illustrate how an integrated platform ties all these capabilities together. GroundHog's mining cloud platform, for example, unifies fleet management, drilling guidance, fuel monitoring, maintenance and control-room workflows into one system. Finally, we offer a practical KPI library of performance metrics, a phased implementation roadmap, change management guidance, case examples, and even a 90-day startup checklist. By the end, you will see how disciplined data and execution unlock both sustainable practices and high mine productivity.

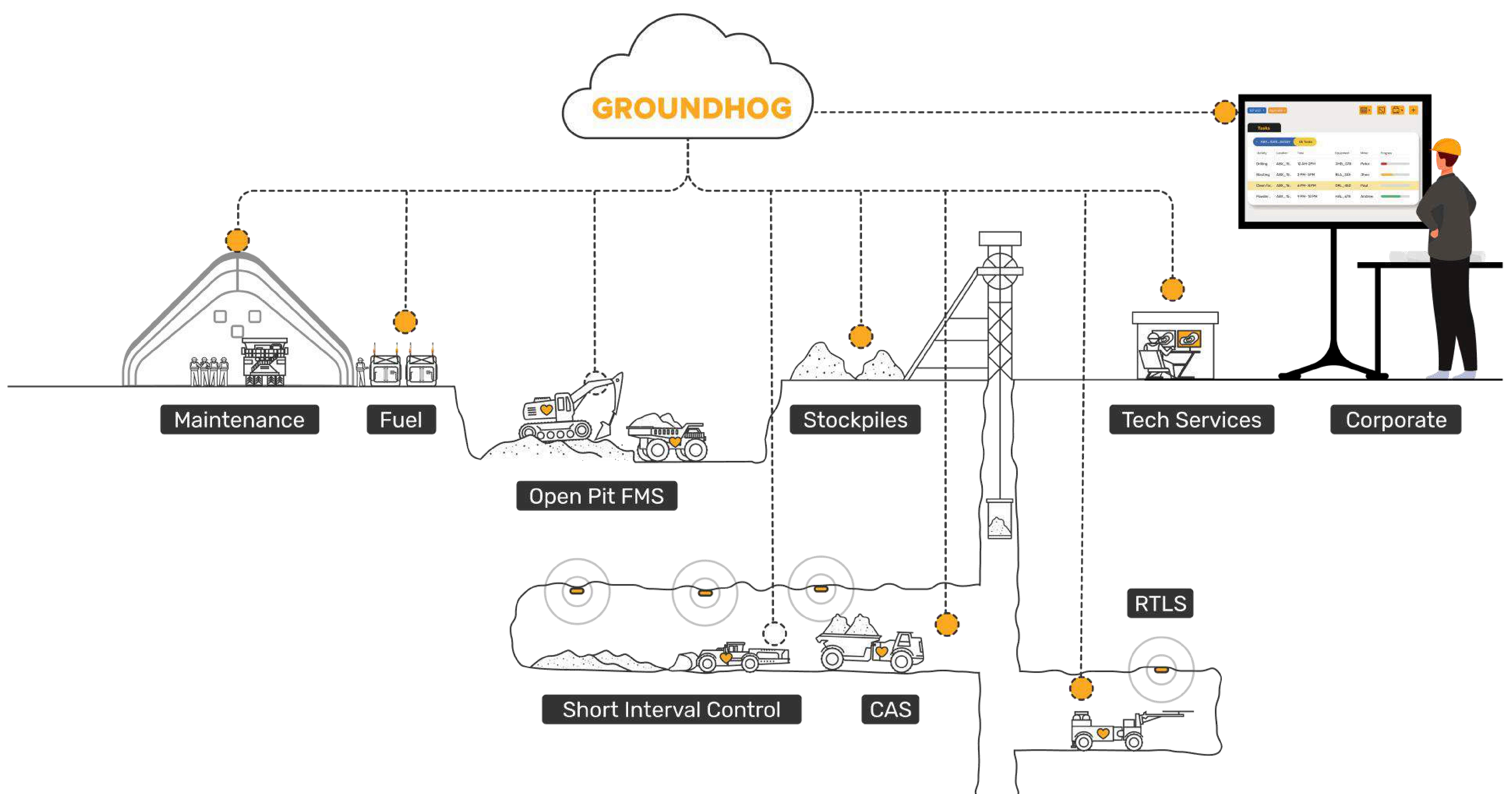
Why Green Mining Is Now an Operational Requirement

Mining today faces an inescapable reality: environmental sustainability is an operational requirement. Regulatory and social pressures have raised the bar. Governments, investors, and communities demand reduced carbon emissions, stricter water management, and better stewardship of land and biodiversity. Mines that fail to adapt risk losing permits, investor funding, and social license. At the same time, the rapid growth of renewable energy, electric vehicles and digital technology is increasing demand for critical minerals, putting pressure on mines to deliver more output.



Key drivers make green mining imperative

- **Connected Equipment and Telematics:** Vehicles, drills, shovels and conveyors are fitted with GPS, payload sensors, and diagnostics. Telematics units on each machine stream location, load and health data. For example, a haul truck might report its load weight and fuel burn every second.
- **Internet of Things (IoT) and Sensors:** Beyond machines, the mine is instrumented with environmental sensors (air quality, dust, gas), geological sensors (vibration, tilt in walls and dams), and wearables for people. These IoT devices send data continuously to the network.
- **Internet of Things (IoT) and Sensors:** Beyond machines, the mine is instrumented with environmental sensors (air quality, dust, gas), geological sensors (vibration, tilt in walls and dams), and wearables for people. These IoT devices send data continuously to the network.



- **Integrated Data Platforms:** All data is aggregated into a central cloud-based platform or data lake. Specialized mine management software integrates production plans, block models, maintenance records and sensor feeds. This provides a single source of truth across departments.
- **Advanced Analytics and AI:** With comprehensive data, mines apply analytics. Dashboards visualize key metrics (tonnage, fuel use, downtime). Predictive models forecast equipment failures or ore grades. Machine learning finds patterns, such as which trucks consume too much fuel on certain routes.
- **Automation and Control:** Some operations become automated. Drills can be guided by software with centimeter accuracy. Autonomous haul trucks follow optimal routes without drivers. Process controls adjust grinding mill speeds based on ore feed rates to save energy.
- **Operational Workflow Tools:** Digital mining also means new ways of working. Supervisors use tablets to distribute tasks and record progress. Dispatchers control fleets through software rather than radio. Geologists update models using mobile apps. Manual forms and paper logs give way to electronic entry and real-time visibility.

In sum, digital mining is an end-to-end approach. It transforms a mine from isolated systems and human guesswork into an integrated, responsive environment. Every drilling plan, blast order, load and haul action is managed, monitored and recorded. Data flows automatically through the organization, enabling a culture of continuous improvement. Ultimately, digital mining provides the infrastructure needed for all the efficiency and sustainability gains that follow.

How Digital Mining Supports Green Mining

"Digital mining" refers to the use of technology like AI, IoT, and automation in the mining process, while "green mining" refers to environmentally friendly practices that reduce a mine's impact, such as using renewable energy and efficient water management. The two concepts are not mutually exclusive; digital mining technologies can be used to achieve green mining goals by increasing efficiency, reducing waste, and enabling better environmental monitoring. Digitalization supports green mining by using technologies like AI, IoT, and automation to increase efficiency, reduce resource consumption, and improve environmental monitoring. Digital tools help optimize water and energy use, minimize waste through better ore sorting, and enable real-time tracking of emissions and other environmental parameters.

Keyways digital mining supports green mining

- **Optimized Resource Extraction:** AI and advanced data analytics help identify optimal drilling patterns and ore grades, allowing miners to extract more valuable minerals with less waste rock. This reduces the amount of material that needs to be processed, which in turn lowers energy and water consumption.
- **Energy Efficiency and Decarbonization:** Digital platforms and IoT sensors optimize energy use in real time, enabling renewables and electric vehicles to reduce fossil fuel use and emissions.

- **Automation and Remote Operations:** Autonomous vehicles, drones, and remote operating centers remove personnel from hazardous areas, improving safety. This technology also enhances accuracy, reduces equipment wear and tears, and allows for operations in remote or challenging environments with higher efficiency and less fuel consumption.
- **Predictive Maintenance:** AI-driven predictive maintenance platforms forecast equipment failures before they happen, allowing for just-in-time repairs. This increases equipment lifespan, reduces the need for spare parts inventories, and minimizes costly and disruptive unplanned downtime, improving overall resource efficiency.
- **Advanced Water Management:** Digital solutions implement closed-loop water systems and advanced filtration, allowing companies to conserve and reuse water and avoid contaminating nearby sources. Real-time monitoring ensures compliance with water quality regulations.
- **Waste and Tailings Management:** Distributed sensor networks and AI models monitor tailings dams in real-time, detecting potential instabilities and preventing catastrophic failures and environmental damage. Digital solutions also enable more efficient processing of waste materials to recover valuable minerals, supporting a circular economy approach.
- **Ecosystem Monitoring and Land Restoration:** Digital tools, such as satellite imaging and AI-based modeling, help conduct rigorous environmental impact assessments and support post-mining land rehabilitation efforts, such as soil stabilization and native vegetation replanting.

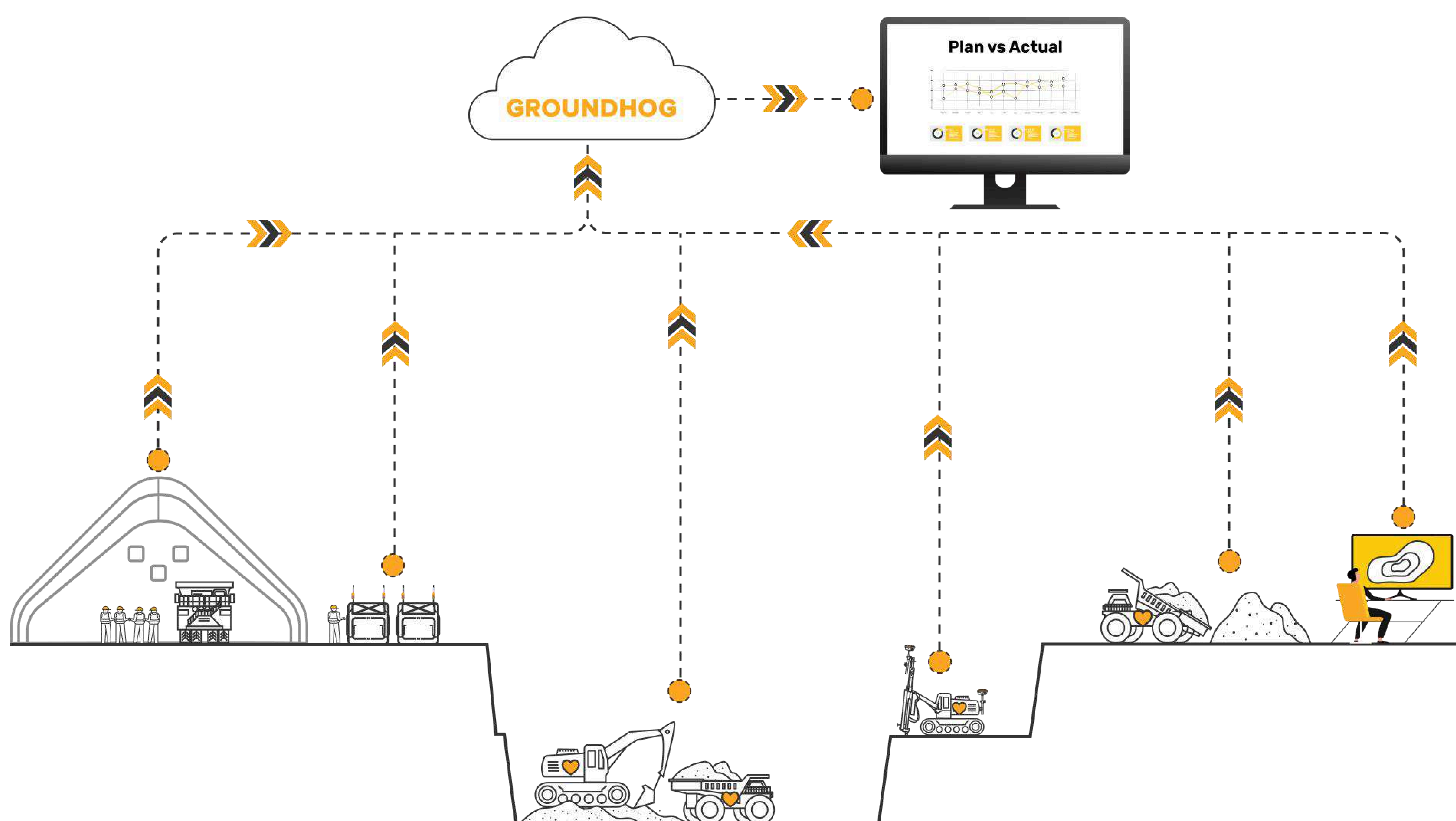
- **Transparency and Compliance:** Digital transformation enables transparent and traceable monitoring of operations through real-time data collection, satellite imaging, and blockchain technology. This allows mining companies to demonstrate compliance with environmental regulations and sustainability-linked financing requirements, building trust with investors and communities.

By leveraging these digital innovations, mining operations can become significantly more efficient, safer, and environmentally responsible, aligning with global demands for sustainable development.

Data Foundations: Capture, Quality, and Context

All digital mining rests on data – and data only matters if it is captured correctly, trusted, and understood in context. Building strong data foundations is the first step

- **Data Capture:** Every relevant asset and process must be instrumented. Equipment is fitted with IoT devices that measure position, speed, payload, and engine status. Pumps, valves and pipelines have flow and level sensors. Even hand-held operations (like inspection of blast holes or loading logs) use mobile apps on tablets to record information. Wherever possible, data is collected automatically: for example, onboard scales log each truck load. Manual entries are minimized to reduce errors.



- **Data Quality:** Raw data can only drive decisions if it's accurate and consistent. This means calibrating sensors, validating inputs, and de-duplicating records. Automated checks should verify data plausibility (e.g. flag a truck reporting 200% of its capacity). Consistent naming conventions and time stamps tie data together. One approach is to assign a data steward to each domain (fleet data, geology data, maintenance data) to oversee quality. Over time, data quality processes ensure that reports and dashboards remain reliable.

Data Context: Numbers gain meaning with context. A ton of ore is just a number unless you know which mine, which bench, and which shift. Digital systems enrich raw data with metadata. Each equipment reading is tagged with equipment ID, operator, and location. Production figures link to blast design codes or conveyor lines. By integrating datasets (for example, connecting a shovel's location to the mine's 3D block model), the platform knows which orebody is being mined. This contextual linking allows managers to ask meaningful questions – for instance, “Which part of the orebody had the highest dilution this week?”

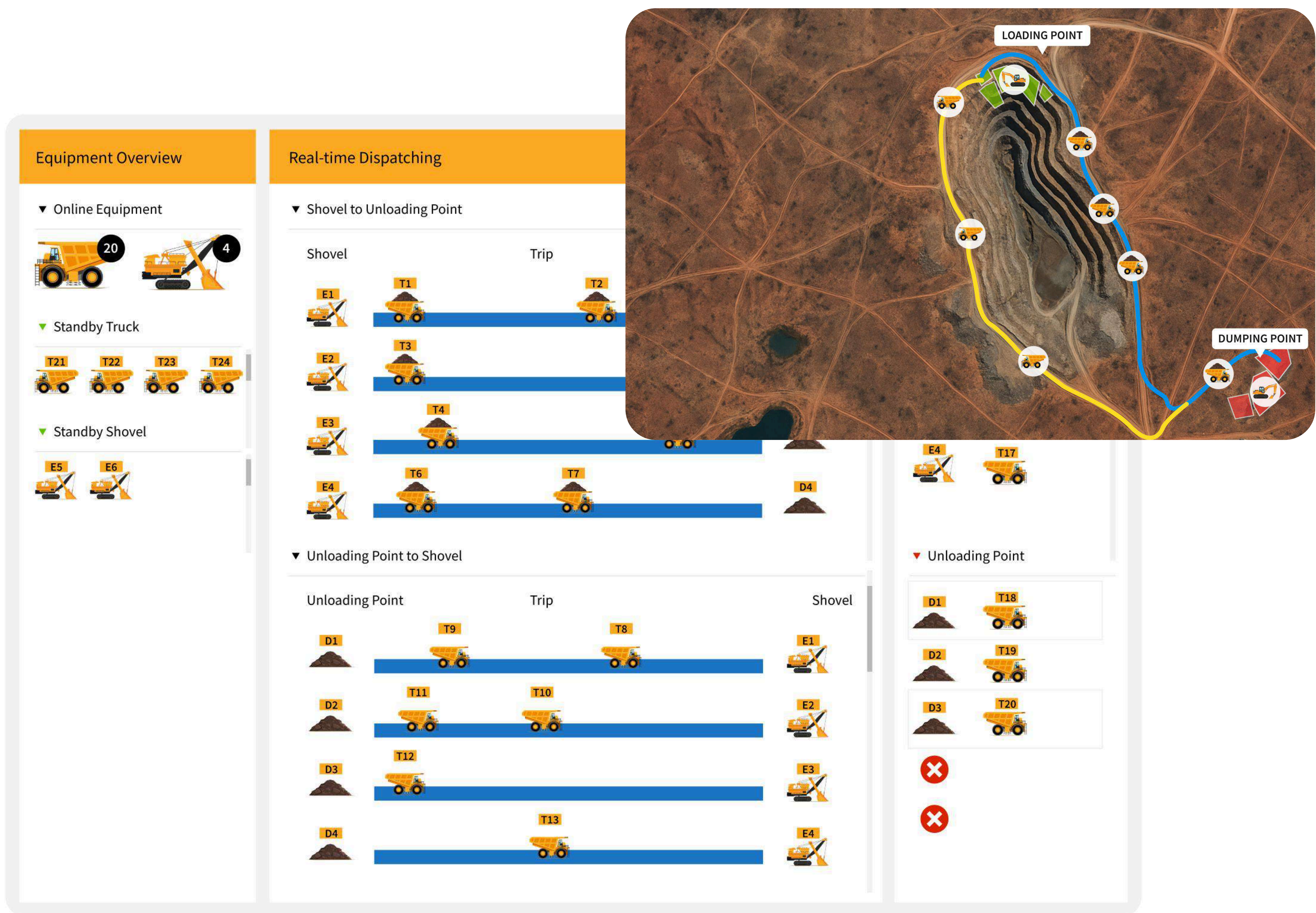
With these foundations, everything else becomes possible. Drill hole locations are entered with exact coordinates and linked to assay results. Haul truck fuel usage is logged with timestamps and road segment labels. Maintenance records tie a failed component to the machine's history. When data capture, quality and context are addressed up front, higher-level applications (analytics, digital twin, etc.) work effectively. Without it, you cannot compare shift to shift or mine to mine.

In practice, building the data foundation means investing in connectivity (Wi-Fi, LTE or satellite), choosing a reliable cloud platform, and training staff on data entry protocols.

Over time, these efforts payoff: managers gain confidence in the numbers and the improvement cycle accelerates. Remember, data is useless if it sits in silos; the goal is a unified, contextual data view of the entire operation.

Fleet, Haulage, and the Fuel-Intensity Lever

A mine’s fleet – trucks, loaders and support vehicles – typically consumes the lion’s share of its energy. Optimizing haulage operations is thus a primary target for both cost savings and emission reductions. In fact, managing fuel intensity has become a strategic lever.



- **Dispatch and Route Optimization:** Digital fleet management systems enable intelligent dispatch. Rather than sending trucks over the radio, dispatch software assigns trucks to loading and dumping locations based on real-time information. The system knows each truck's location, fuel state, and assigned task. It calculates the most efficient routes and queues to minimize empty travel and idle time. For example, it can schedule a truck to pick up the next load before it reaches the dump site. By optimizing truck matching and routing, the mine reduces fuel burned and equipment wear.
- **Real-Time Fuel Monitoring:** Equipping vehicles with fuel level sensors allows minute-by-minute tracking of consumption. The platform aggregates this data to show where and when fuel is used. Advanced analytics can detect anomalies – say, an unexpected fuel spike on a route – and alert managers. It can also correlate fuel use with load weight and gradient. Some systems automatically compute fuel burn per loaded tonne on each ramp or road segment. Armed with this insight, operations can redesign ramps, adjust loads, or change driving practices to save fuel.
- **Payload Optimization:** Every hauling cycle's efficiency depends on payload. Overloaded trucks strain engines and use extra fuel; underloaded trucks mean energy is wasted moving dead weight. Digital payload monitoring (using onboard scales or weight estimation systems) ensures each truck runs at its optimal load. When an operator loads a bucket, the system records it; if a truck consistently runs below target weight, dispatch can adjust assignments. Over time, maintaining a high average payload boosts productivity and stabilizes fuel consumption.

- **Fuel Theft and Leak Prevention:** Another benefit of sensor monitoring is security. Sudden drops in recorded fuel levels can indicate leaks or theft. An intelligent system cross-references fuel data with ignition status and location to identify suspicious events. Quick alerts allow supervisors to investigate – preventing large cumulative losses that also harm productivity.
- **Electrification and Alternative Fuels:** Many mines are now piloting electric or hybrid haul vehicles. These reduce carbon output directly. Digital systems still play a role in managing them. For electric trucks, monitoring state-of-charge and scheduling charging is essential. Smart dispatches ensure vehicles recharge during off-peak hours or when renewable power is available. The platform treats electricity similarly to diesel: it tracks kWh per tonne and integrates renewable energy forecasts to minimize carbon intensity.

- 1. Optimize haul routes and scheduling through digital dispatch systems.**
- 2. Monitor fuel use in real time with onboard sensors.**
- 3. Use analytics to identify and eliminate fuel “hotspots” on ramps and roads.**
- 4. Prevent fuel waste by alerting on theft or leaks.**
- 5. Automate payload weighing keep trucks optimally loaded.**
- 6. Manage electrification or alternative fuels via smart charging and usage tracking.**

By focusing on the haulage fleet, a mine makes the biggest dent in its fuel consumption. Each liter saved is a direct emission cut. The disciplined use of fleet data translates immediately into environmental

benefit (fewer CO₂, NO_x and particulate emissions) and into the bottom line (lower fuel expenses). In practice, mines often see double-digit improvements in fuel efficiency over a year of disciplined fleet management, making this one of the highest-impact digital initiatives.

Grade Control, Drilling Precision, and Waste Reduction

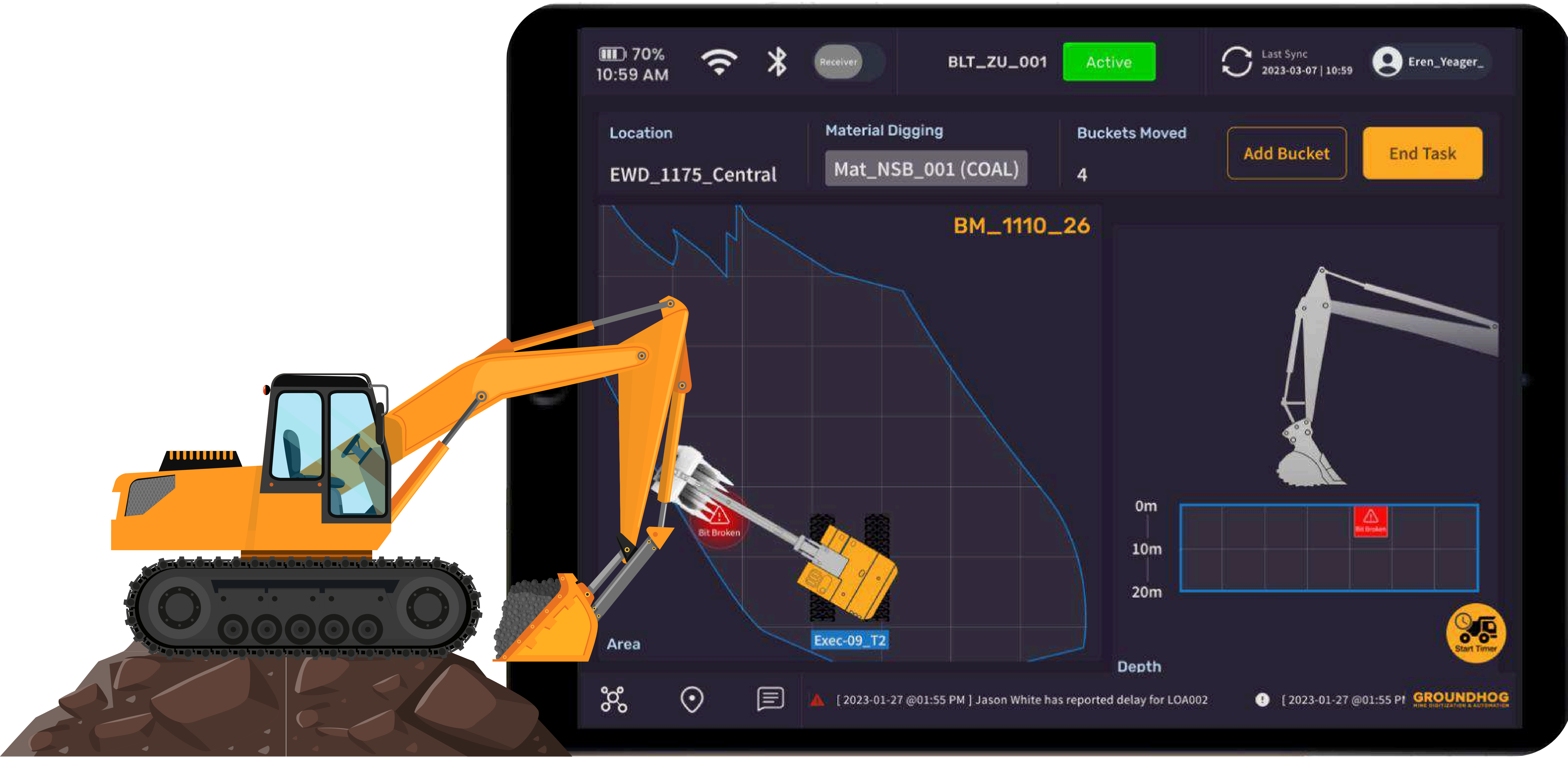
Another key lever is the material that is moved. When mines hit or miss the ore/waste boundaries inaccurately, they incur cost and environmental penalties. Digital grade control and precision drilling reduce waste rock and maximize ore recovery.

- **High-Precision Drilling:** Modern drilling rigs equipped with GPS and guidance apps can drill blast holes with centimeter accuracy. The plan is uploaded to the drill operator's tablet, and the rig's movements are guided in real time. This ensures holes align with the intended pattern and depth. The immediate benefit is precise controlled drilling at Ore Contact Zone reduces ore dilution. It also enables better fragmentation – precisely drilled blasts shatter rock into more uniform pieces, reducing oversize boulders. That means loaders and crushers spend less energy breaking rock into mill feed.
- **Digital Grade Control:** Grade control is the process of separating ore from waste rock. Traditionally, crews would dig benches and then rely on periodic sampling to update models. Digital systems accelerate this. Surveyors and geologists use mobile apps to log drill-hole data and shovel samples in real time. The digital platform updates the orebody

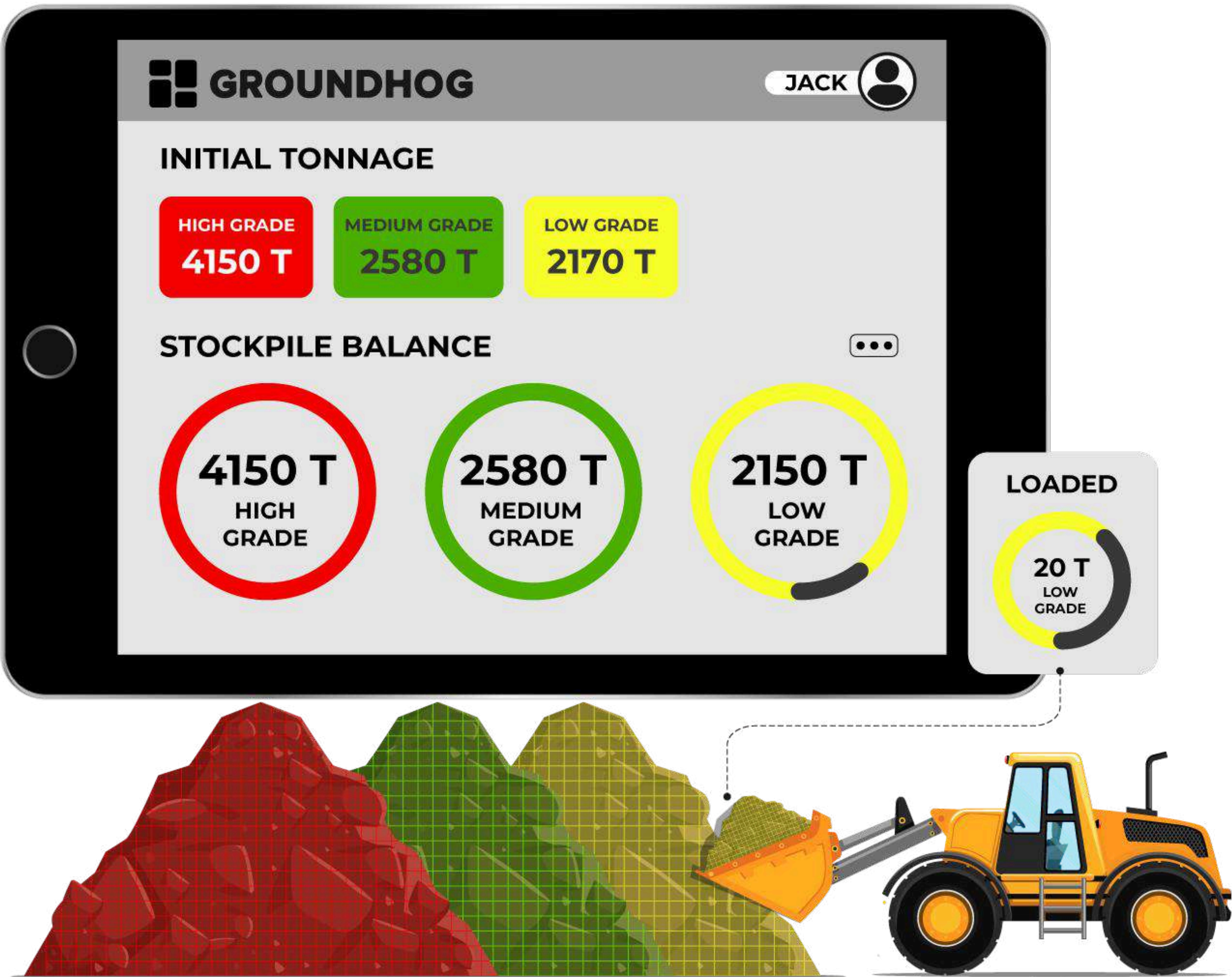
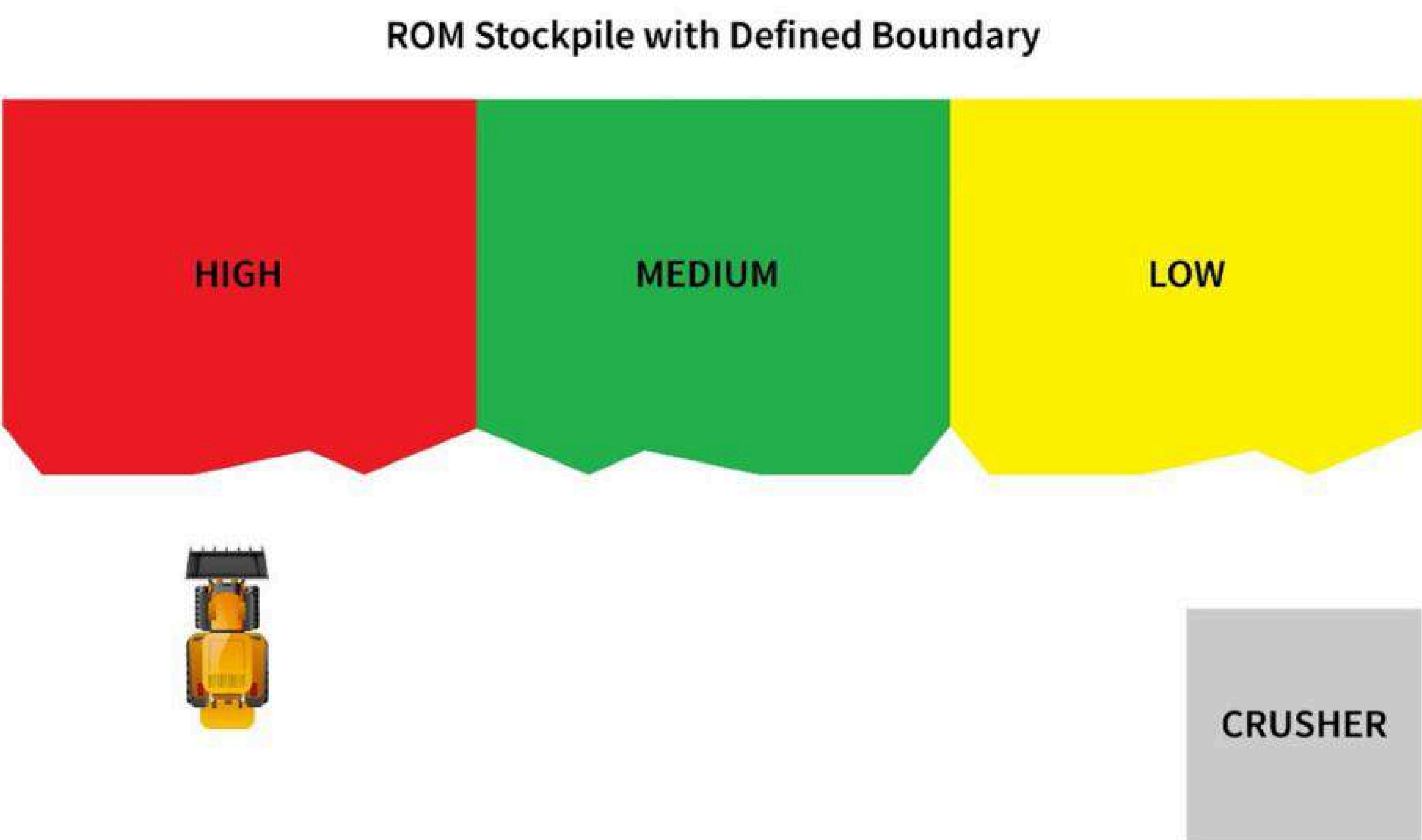
soon as new information arrives. The result is a moving window into the ore zone. When the model shifts, mining plans can be adjusted within hours, minimizing the chance that valuable ore is left behind or too much waste is sent forward.



- High Precision Digging and Deviation Alerts:** High Precision Digging and Deviation Alerts: Integrated platforms can automatically detect when operations deviate from plan. If an excavator digs outside the planned ore zone, the system flags the error immediately, allowing the crew to correct course on the very next pass. Over time, operators learn to follow the updated model more closely, improving consistency and accuracy. Some systems even provide a visual overlay on equipment monitors, showing exactly where and how deep to dig. This real-time feedback loop tightens execution, reduces ore dilution, minimizes rework, and significantly improves overall productivity and resource recovery.



- **Ore Sorting and Quality Monitoring:** Some advanced operations add ore-sorting sensors at the pit. For example, hyperspectral or XRF analyzers on loaders can analyze each bucket’s grade before it is dumped. If the mix contains too much waste, that bucket can be diverted to a waste pile. Conversely, higher-grade buckets go to the mill. This real-time grade sorting can increase overall plant feed grade without moving extra rock. Even without expensive sorters, simply tracking grade mix on each load helps managers spot dilution issues quickly.



- 1. GPS-guided drilling for exact hole placement, improving fragmentation and reducing re-drills.**
- 2. Digital drilling plans and drill app synchronization, ensuring operators hit designated targets.**
- 3. Real-time grade control: rapidly log and process field assays to delineate ore boundaries accurately.**
- 4. Automated deviation alerts: notify excavator operators if they mine outside the intended ore zone.**
- 5. Loader payload tracking: confirm each load's expected material mix.**
- 6. AI-grade modeling: refine ore/waste estimates as new data comes in, minimizing dilution.**

The upshot is less waste rock excavated, and less ore accidentally left behind. Cutting waste directly saves fuel and processing costs (you're not hauling unnecessary rocks). It also preserves more reserves for future recovery, effectively extending the mine's productive life. In other words, precision in drilling and loading is a form of waste reduction that yields both economic and green dividends.

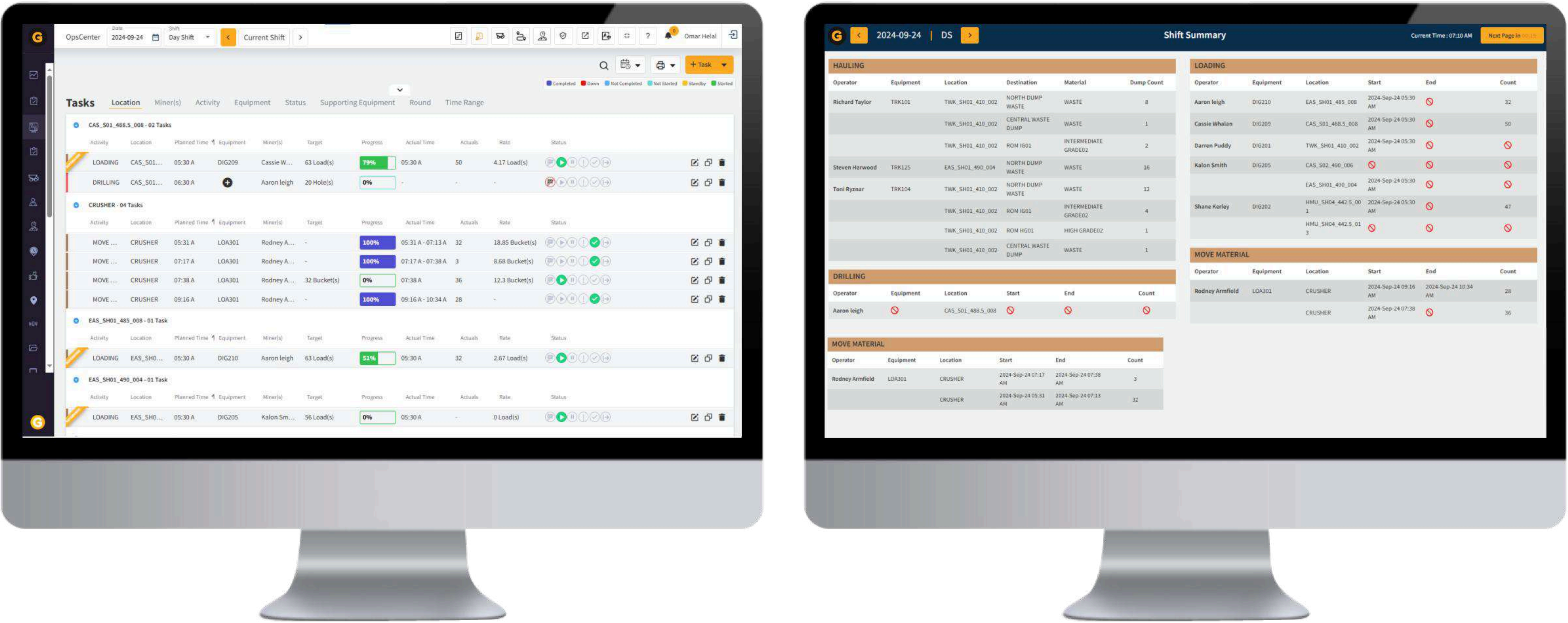
Short Interval Control and Execution Discipline

Having great plans and data is one thing; execution is another. Short Interval Control (SIC) ensures that plans are followed closely on the floor, and any deviations are corrected on the flight. Digital tools make SIC practical and powerful.

SIC breaks the shift into small intervals (hours) and actively manages tasks throughout the shift. Without digital support, this is tedious and error prone. With a modern platform, everything syncs.



- **Task Planning and Allocation:** In a digital SIC system, a mine's long-term schedule is imported, and daily or shift tasks are automatically generated. The software assigns each task to specific crews and equipment, considering training and availability. For example, if three shovels, five trucks and two drills are needed in the morning, the system can pre-allocate them with one click. This replaces manual lists or radio assignments, speeding up shift starts and reducing confusion.
- **Real-Time Progress Tracking:** As work proceeds, data is captured continuously. An operator marks each completed bucket or haul, or the equipment's telemetry automatically logs each load. The SIC dashboard shows tons delivered vs. plan, equipment statuses, and fuel or explosives used. Supervisors monitor this live feed. If a team is falling behind, they get alerted immediately and can reassign resources or adjust priorities on the spot.



- **Communication and Visibility:** Platforms like GroundHog allow crews and supervisors to see tasks on tablets or wall-mounted screens. Each worker knows their tasks and deadlines. Supervisors use an app to distribute new tasks or update instructions. This clear communication means fewer misunderstandings. For example, if geology sends a revised blast design mid-shift, it can be pushed to the drillers digitally without interrupting work.
- **Issue Resolution:** Any delay or problem is logged instantly. If a truck breaks down, that downtime is entered (manually or automatically), and the system reallocates remaining work. By the end of the shift, the system has a detailed report: tons moved by area, hours lost to downtime, fuel consumed, etc. These reports are reviewed to identify lessons (for instance, a recurring backlog at a crusher might indicate a needed improvement).

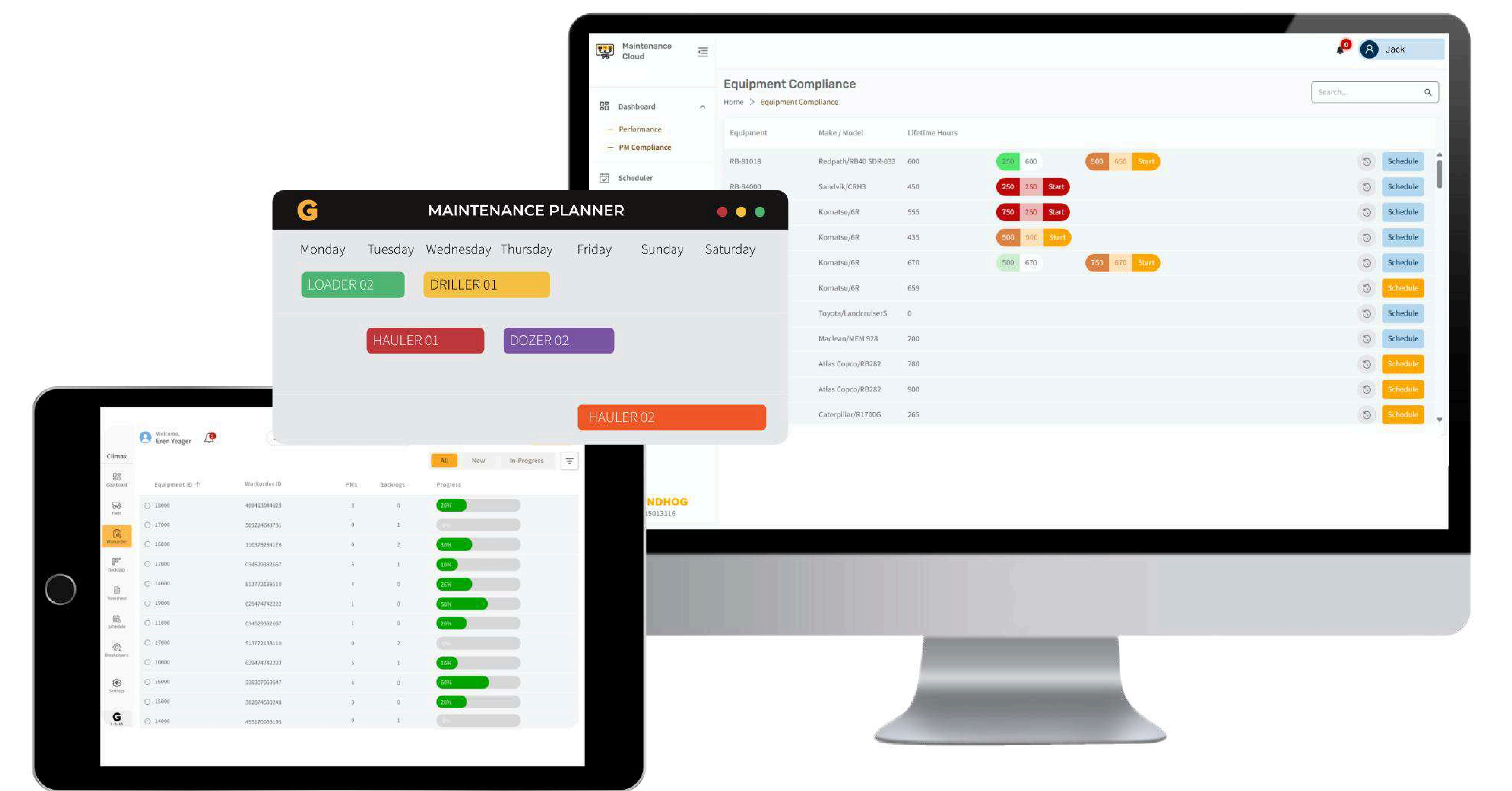
- 1. Automated shift task allocation: crews and machines assigned to tasks in the system for quick shift starts.**
- 2. Live production vs. plan dashboards: supervisors compare actual tons and progress to the schedule in real time.**
- 3. Clear task communication: tasks and priorities sent to tablets, reducing confusion and downtime.**
- 4. In-shift adjustments: the team can re-prioritize or reassign resources immediately when deviations occur.**
- 5. Detailed shift reports: at shift end, automated logs show KPIs (production, delays, maintenance issues) for analysis.**

The effect of disciplined short-interval control is a culture of accountability. Workers see exactly what was planned, and management sees exactly what happened. Productivity improves because small setbacks don't snowball; they are caught early. This discipline also affects green metrics. For example, a truck that is behind schedule might idle more; in a SIC system, that idle time is visible and can be cut. Fuel is only burned for productive work. Ultimately, SIC with digital tools aligns daily execution tightly to both operational and environmental goals.

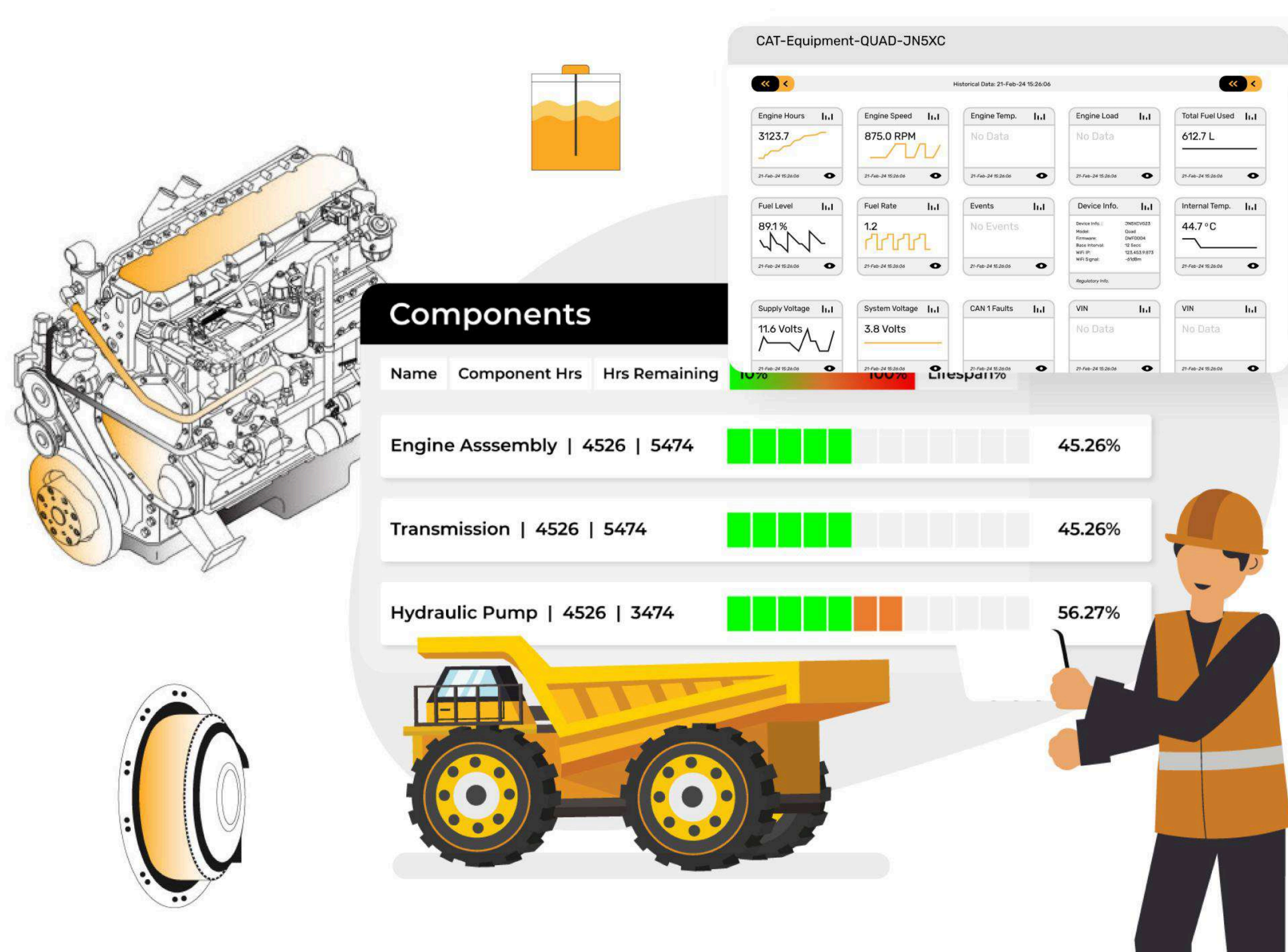
Maintenance, Reliability, and Resource Life

Equipment downtime is wasted opportunity. Every hour a truck or shovel is idle due to breakdown is an hour of production lost (and often diesel idled or wasted). Digital maintenance management keeps machinery healthy, which conserves resources and extends asset life.

- **Mobile CMMS:** Digital maintenance platforms (often cloud-based) allow field technicians to use tablets to view and update work orders. Instead of paper logs, an operator can log a breakdown on the spot. The system then automatically notifies the maintenance planner, who schedules a technician. Once the repair is done, the tech records it via the app. This mobile workflow eliminates delays and errors in communication.



- Predictive and Condition-Based Maintenance:** Beyond logging problems, mines use sensors to predict them. Engines, gearboxes and hydraulics are fitted with monitors. Vibration, oil quality, and temperature sensors feed into predictive models. If a sensor indicates a part is wearing abnormally, maintenance is scheduled before a catastrophic failure. This approach increases machine availability and avoids emergency shutdowns, which are both costly and dangerous.



- Spare Parts and Inventory:** A digital system tracks spare parts inventory. It can automatically reorder critical parts when stock is low. This prevents situations where a machine sits broken for days waiting on a filter or sensor. Efficient inventory management reduces capital tied up in spares while ensuring reliability.
- Maintenance Analytics:** With all maintenance data in one place, analysts can spot trends. Which trucks are failing most often? Are certain failure modes happening at a particular age? This reliability analysis can feed back into operations (e.g. rotating trucks through different routes) or

or design (e.g. buying higher-spec parts).

- 1. Implement a computerized maintenance management system (CMMS) with mobile access.**
- 2. Install sensors for condition monitoring (vibration, oil, temperature) to enable predictive maintenance.**
- 3. Track maintenance backlog and completion rates; aim to clear critical work orders on time.**
- 4. Schedule preventive maintenance based on usage and condition (not just fixed intervals).**
- 5. Optimize spare parts inventory with real-time tracking and reorder alerts.**
- 6. Analyze failure data to extend asset life and improve maintenance plans.**

Better maintenance means higher equipment availability and longer life for each machine. That translates into more consistent production (fewer unscheduled stops) and defers the need to buy new equipment. In green terms, reliable machines burn fuel more efficiently and avoid extra consumption that often occurs during breakdowns. Moreover, extending the life of heavy equipment distributes its embedded carbon footprint over more years of service. In short, digital maintenance practices protect both production and sustainability performance.

Water, Tailings, and Environmental Controls

Mining's impact extends beyond rocks: water use, tailings disposal, and environmental emissions are critical concerns. Digital solutions now help mines monitor and manage these aspects tightly.

- **Water Management:** Sensors measure flow rates, storage levels, and water quality throughout the site (from process tanks to sumps to tailings ponds). Advanced control systems use this data to recycle water efficiently. For example, if a sensor shows the thickener underflow is too diluted, the system may adjust flocculant dosing to recover more water. In some mines, central control software ensures that pumps run only when necessary, preventing water losses. On the user side, dashboards track total water consumption and flag any abnormal usage.
- **Tailings Monitoring:** Tailings dams and storage facilities are fitted with geotechnical sensors (piezometers, inclinometers, strain gauges, etc.) that feed a monitoring system. If a sensor detects abnormal pore pressure or movement, alerts are issued immediately to inspect the dam. Some mines create a digital twin of their tailing's impoundment. By inputting real sensor data into a model, engineers can simulate future scenarios (such as heavy rain or seismic events) to plan preventive measures. This proactive approach minimizes the risk of failures and uncontrolled spills.
- **Emission and Dust Control:** Weather stations and dust monitors track air quality around the site. If particulate levels rise, the system can automatically activate dust suppression measures (sprinklers or foggers) on roads and conveyors. For diesel-powered equipment, the digital log

of operating hours and fuel use can estimate NO_x emissions; some sites also use catalytic reduction monitored by software to ensure engine emissions stay within limits. Even greenhouse gases (CO₂) can be monitored if the mine's carbon accounting is integrated with its digital platform.

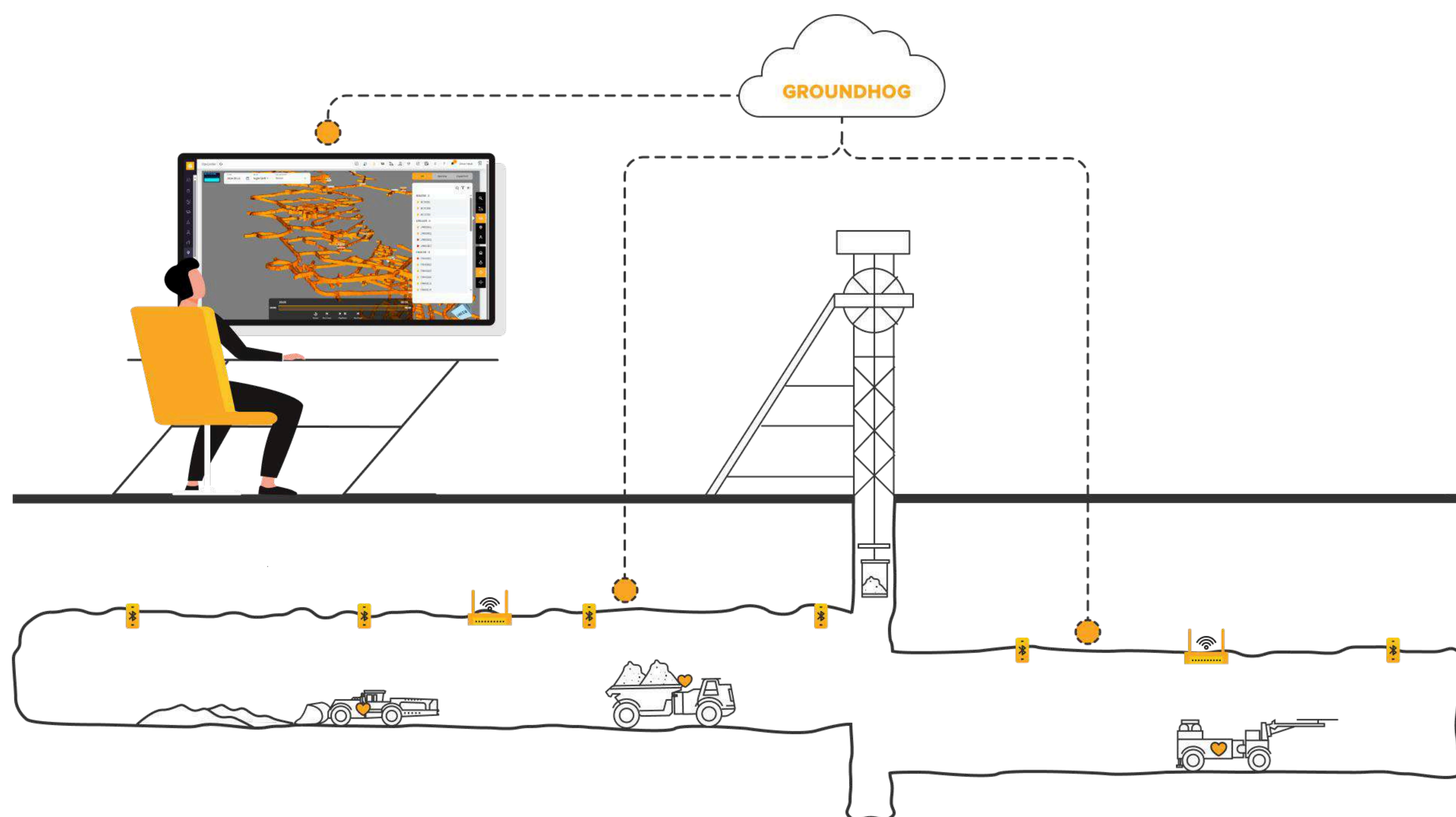
- **Biodiversity and Rehabilitation:** Modern sites may use drones and imaging to map vegetation and terrain changes. GIS and sensor data can guide progressive rehabilitation – for example, ensuring that returned waste rock is contoured correctly and seeded in line with climate conditions. Soil and moisture sensors in revegetation areas provide feedback to ensure that rehabilitation is taking hold as intended.
 - 1. Deploy IoT sensors to monitor water levels, flow rates and quality throughout the circuit.**
 - 2. Automate pump and valve controls to maximize recycling and minimize discharge.**
 - 3. Use digital models and sensors to monitor tailings dam stability and optimize dewatering.**
 - 4. Install air-quality and dust monitors; trigger dust suppression or reduced activity as needed.**
 - 5. Track energy use (electricity vs diesel) to assess greenhouse gas output.**
 - 6. Map and monitor rehabilitation areas with drones and GIS tools.**

Digital environmental control turns open-loop processes into closed-loop systems. Instead of routine schedules (like pumping at a fixed rate), the operation adjusts in real time to data. This leads to safer, compliant conditions with minimal resource use. For instance, a properly controlled

tailings thickener might save millions of liters of water per year by optimizing sedimentation. Each technology that cuts water or energy use has immediate payoff, and the data proving those savings can be used in sustainability reporting.

Sensors, RTLS, and Worker Health Monitoring

Mining's digital transformation includes the people on site. Sensors and tracking systems protect workers and enable better health monitoring, tying into the social and safety aspects of sustainability.



- **Wearable Health Sensors:** Wearables like smart helmets or wristbands monitor vital signs (heart rate, body temperature, hydration) in real time. In hot or high-altitude mines, these can detect fatigue or heat stress before it becomes dangerous. If a miner's heart rate or core temperature exceeds safe levels, the system can alert supervisors, who

can prompt a break or medical check. Over time, aggregated data can optimize work-rest schedules and medical support, improving well-being and productivity.

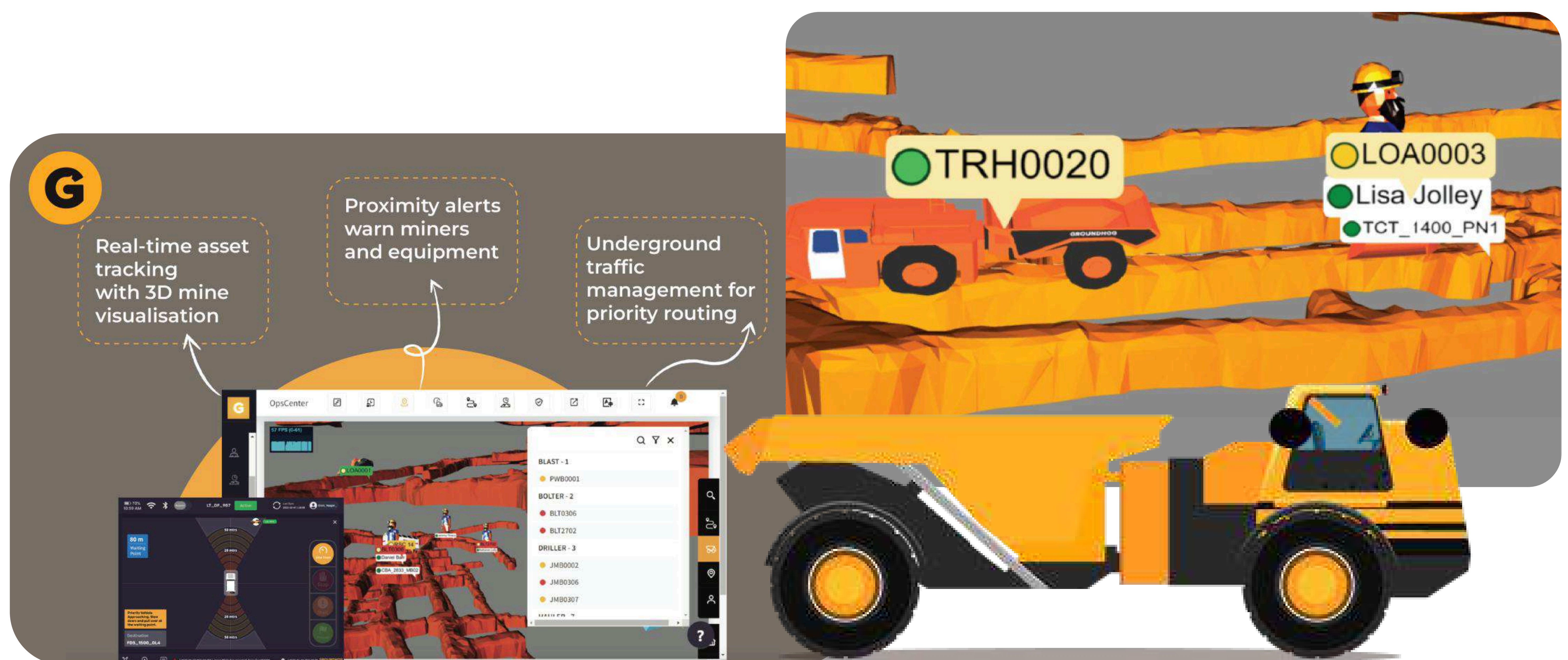
- **Real-Time Location Systems (RTLS):** Technologies such as GPS, RFID or ultra-wideband track the location of personnel and mobile assets. Outdoors GPS can show teams in the pit; underground, UWB tags or leaky feeder systems pinpoint workers. RTLS is used for two main purposes: safety and efficiency. On safety, if a sensor signals a hazard (such as a rock fall or gas leak), the system automatically identifies and alerts any workers in that zone. In emergency evacuation drills, RTLS confirms everyone is accounted for. On efficiency, knowing exactly where crews are can improve dispatching and ensure faster responses.



- **Environmental Hazard Sensors:** Personal gas detectors and area sensors continuously measure conditions like methane, carbon monoxide or oxygen levels, especially underground. These sensors connect to the network; if levels exceed safe limits, alarms are triggered and work can be halted automatically. Similarly,

sensors monitor noise or vibration levels that could affect worker health, ensuring the environment stays within safe thresholds.

- 1. Wearables: Equip workers with devices that monitor vital signs and send alerts if readings go unsafe.**
- 2. Location Tracking: Use GPS/RTLS to know where people and vehicles are, enabling quick response to incidents.**
- 3. Environmental Sensors: Deploy gas detectors, air quality monitors and noise sensors as needed.**
- 4. Geofencing: Set up virtual boundaries – e.g., around active blasts or highwalls – that generate warnings if entered.**
- 5. Data Integration: Feed all safety data into the same platform so alerts and insights are unified with operations data.**



When these systems are integrated, mines reach a higher level of safety. Fewer accidents mean less downtime, and a healthier workforce means fewer disruptions. From a sustainability view, this strengthens the social pillar of ESG—no mine is truly “sustainable” if it endangers people. Digitally enhanced safety also brings efficiency gains, such as smarter ventilation that saves energy while improving air quality.

Automation, Robotics, and Safer Green Outcomes

Automation is a natural extension of digital mining. It shifts routine or dangerous tasks to machines, which operate more consistently and safely. In doing so, automation often reduces waste and energy use, aligning perfectly with green goals.

- **Autonomous Haulage:** Mines worldwide deploy driverless haul trucks. These trucks follow optimized routes and schedules, operating up to 24/7 without human breaks. They maintain consistent speeds and follow traffic rules precisely, which improves fuel efficiency. Without driver fatigue, accidents are reduced. For electric or trolley-assisted trucks, autonomy means maximizing uptime on electric power. The precision of autonomous dispatch also avoids empty trips. Overall, automating haulage has led many operations to higher productivity and lower fuel use.



- **Remote Operation of Equipment:** Some equipment, such as drills, bulldozers, or underground loaders, can be tele-operated from a safe control center. Operators use joysticks or VR controls to run the machines. This keeps people out of harm's way (blasts, rockfall, dust). It also smooths operation: machines can run continuously under stable control. For example, a drill rig under remote control can operate steadily overnight, achieving better penetration rates.
- **Drones and Robotic Inspections:** Routine surveys and inspections are increasingly done by drones or robots. Drones carry cameras and LiDAR scanners to map the pit floor and tailings dams. They gather survey data in hours that would take days on foot. This continuous monitoring allows quicker detection of issues or opportunities for optimization. Robotic crawlers or fixed scanners can inspect conveyors and crushers in real time, detecting wear or misalignment before breakdown.
- **Process Automation:** Inside plants, conveyors, crushers and sorting machines can be automated to run at optimal speeds without needing constant adjustments by operators. Sensors monitor flow and quality, and control systems tweak motor speeds or feeder rates as needed. This keeps equipment running at peak efficiency.
 - **Deploy autonomous haul trucks and loaders to optimize cycle times and save fuel.**
 - **Implement remote or automated drills and dozers to increase accuracy and uptime.**
 - **Use drones and robots for routine mapping, inspections and maintenance checks.**
 - **Automate conveyors and processing for efficiency.**

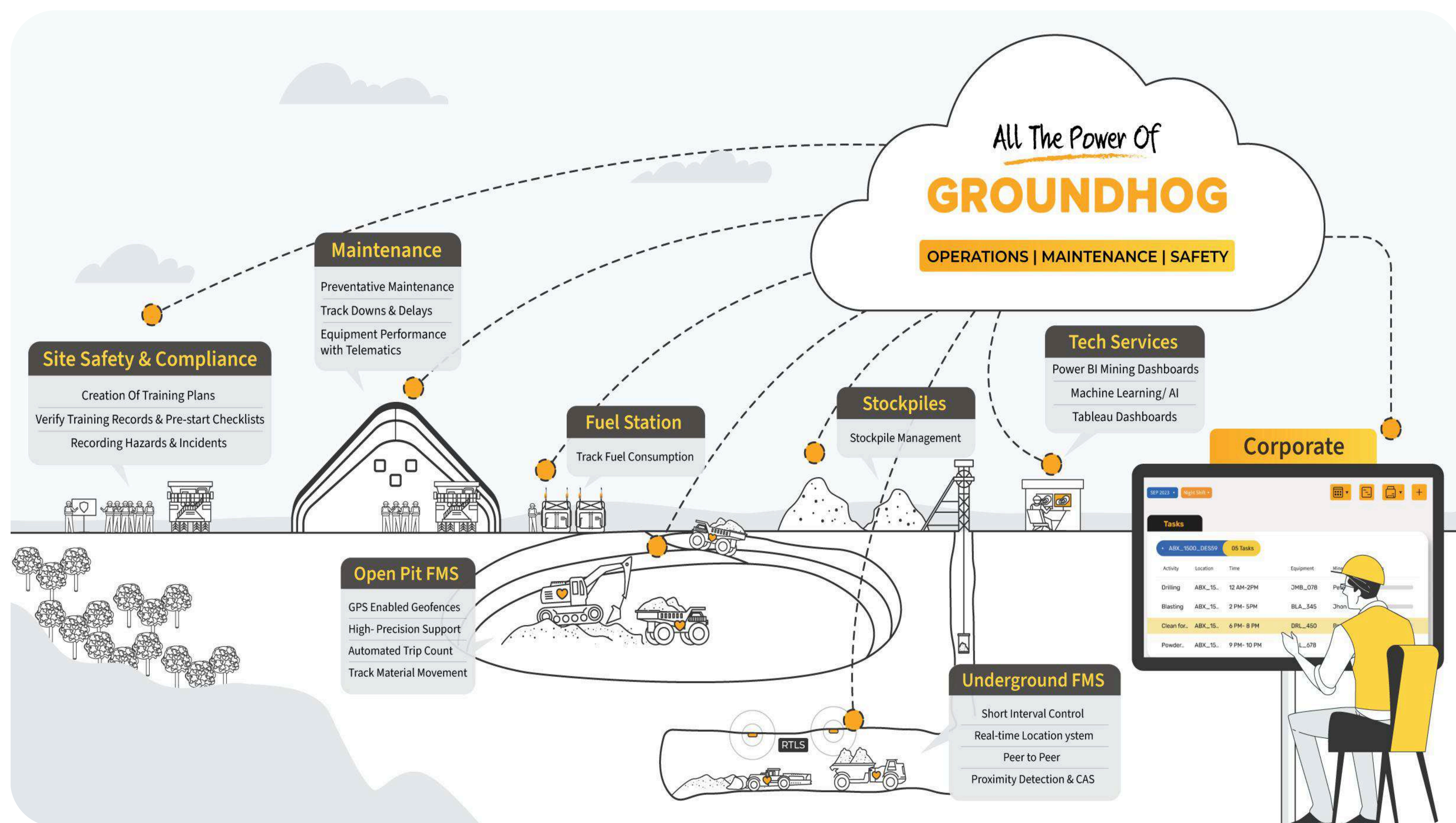
The result of automation is twofold: safety improves because fewer people are exposed to hazards, and sustainability improves because processes run more smoothly with less waste. Machines don't waste time – if a drill finishes a row, it can automatically move to the next task. That means less non-productive idling. In many cases, automation has led to double-digit productivity gains. In a world where environmental impact is often proportional to operating hours, getting more work done per machine-hour directly reduces the environmental footprint per unit of production.

Digital Twins, Cloud Analytics, and Cross-Site Benchmarking

With data flowing from all corners of the operation, mining companies can create a virtual model of the entire enterprise: a digital twin. This, combined with cloud analytics and cross-site comparison, drives strategic insights.

- **Digital Twin:** A digital twin is a real-time, virtual replica of mine. It integrates the 3D geology model, mine design, equipment models and live data feeds. For example, as a dragline works on the face, the twin shows how the bench profile changes. Engineers can use the twin to run “what-if” simulations before making changes. Suppose planners want to alter the pit sequencing to extract a higher-grade zone first; they can simulate that scenario in the twin to predict effects on inventory, energy use, and tonnage.
- **Cloud Analytics:** All the collected data resides in the cloud, where powerful servers run analytics. Data scientists can apply machine learning algorithms to historical data to improve forecasts – for instance

predicting crusher feed fluctuations based on radar stockpile sensors, or recommending optimal blasting parameters. Cloud platforms also provide interactive dashboards that managers can access anywhere. These might display KPI trends, heat maps of truck activity, or breakdown forecasts.



- Cross-Site Benchmarking:** Companies with multiple mines gain from comparing performance. If Mine A has a lower fuel intensity or higher equipment availability than Mine B, management can investigate why. Are the drill rigs newer? Is the haul road design better? Lessons from one site can be adopted by others. Cloud platforms enable this by unifying data across operations. Metrics like “ore tons per liter” or “water use per ton” are reported uniformly, making apple-to-apple comparison possible. Mines can then share best practices, creating a data-driven culture of competition and improvement.

- 1. Build a real-time digital twin of mine and plant to test changes virtually.**
- 2. Leverage cloud computing to analyze big data, apply machine learning and host dashboards.**

- 3. Benchmark across sites: compare KPIs like fuel use or unit costs between mines to identify best practices.**
- 4. Collaborate via the cloud: project updates, alerts and analytics are shared enterprise wide.**

These integrated technologies turn data into foresight. Rather than reacting to yesterday's figures, the team can predict issues and opportunities. For example, if analytics forecast a drop in ore grade in two weeks based on current geology, planners can adjust recovery strategies now to maintain output. As mines become more digital, decisions become more centralized and strategic. The cross-site perspective also ensures that sustainability improvements are scaled: a clever energy-saving tweak found at one mine can be implemented at others.

How GroundHog Ties It Together – Platform Capabilities and Workflows

All the above requires an integrated platform. GroundHog's mining cloud is one example of a unified solution that brings together fleet, fuel, planning, maintenance and safety into coherent workflows.

GroundHog provides both a cloud-based OpsCenter (dashboard) and a suite of mobile apps on tablets or rugged devices in the field. It covers key areas:

- **Short Interval Control:** Manages daily tasks and crews in the system, generates shift reports and displays production vs plan continuously.
- **Fleet & FMS:** Tracks every vehicle's location, status and payload. Shows KPIs like tons per hour and fuel burn in real time.

- **High-Precision Drilling:** Guides drill rigs via GPS to hit planned holes with centimeter accuracy and logs actual holes.
- **High-Precision Digging:** Guides loading equipment via GPS to load planned dig blocks with centimeter accuracy and precise excavation.
- **Maintenance:** Mobile work orders let operators report breakdowns instantly. Maintenance Managers assign tasks and monitor completion.
- **Fuel Management:** Reads fuel sensors on each machine, maps consumption to trucks and routes, and detects inefficiencies or theft.
- **Safety & RTLS:** (If implemented) tracks people and sends alerts if hazards are detected.



What makes an integrated platform valuable is the data flow between modules. For example, a loading shovel's position and payload are sent to OpsCenter and update the SIC progress. When that material moves to the crusher, the tons are fed into the fleet report for trucks to haul. If a truck reports a fuel spike, supervisors can pull its recent assignments and see if it was idling or running uphill.

Key platform workflows include:

- **Data Unification:** Collect and unify telemetry, operational and maintenance data in one platform.
- **Integrated Planning:** Import mine plans (from Deswik or spreadsheets) and distribute tasks through the platform.
- **Field Apps:** Equip crews with mobile apps for drilling, dispatching, and reporting tasks.
- **Real-Time Dashboarding:** Use the OpsCenter to visualize KPIs (production, fuel, etc.) for engineers and managers.
- **Alerts and Analytics:** Monitor performance metrics and send alerts (e.g. fuel used above a threshold).
- **Continuous Improvement:** Use built-in analytics (like fuel hotspot detection or drill compliance reports) to inform next steps.

GroundHog is designed to illustrate how a single system can replace silos. With this approach, none of the operational data is locked away; instead, everything from truck loads to maintenance logs is in one place. This makes green and operational management a unified exercise: for example, you can correlate a spike in fuel burn (from the Fuel module) with a delay in blasting (from the SIC module) to see how inefficiency in one area affects another.

While GroundHog is just one platform among others, its comprehensive suite shows how integrated software enables a mine to move from scattered data to coordinated action. In the final sections on metrics, cases and implementation, keep in mind that such a platform is the enabler for turning data into discipline and discipline into green performance.

Metrics That Matter – A Practical KPI Library for Green Mining

To measure progress, mines track both productivity and sustainability indicators. Below is a sample of key metrics (with formulas) that mines commonly use:

- **Ore Production (tonnes):** Total ore mined or processed per day/shift. (Monitors output volume.)
- **Production Cycle Time (minutes):** Total time for one haul cycle (load, haul, dump) / number of cycles. (Shorter is better; indicates fleet efficiency.)
- **Equipment Availability (%):** $(\text{Operating Time} \div \text{Scheduled Time}) \times 100$. (Fraction of scheduled time machines were running.)
- **Overall Equipment Effectiveness (OEE):** Availability $\times$ Performance $\times$ Quality. (A composite measure of machine productivity.)
- **Ore Recovery Rate (%):** $(\text{Ore to mill} \div \text{Ore planned}) \times 100$. (Shows how much planned ore was recovered.)
- **Dilution (%):** $(\text{Waste in ore} \div \text{Total material to mill}) \times 100$. (Lower is better – it means less waste ground.)
- **Fuel Intensity (L/tonne):** Total diesel consumed (L) $\div$ Total ore hauled (tonnes). (Key sustainability metric – how much fuel per output.)
- **Energy Intensity (kWh/tonne):** (Total energy (kWh, including diesel equivalence) $\div$ Ore processed (tonnes)).
- **Maintenance Backlog:** Number of overdue work orders or % of maintenance tasks completed on schedule. (Signals reliability.)
- **SIC Task Compliance (%):** $(\text{Short-interval tasks completed as planned} \div \text{tasks planned}) \times 100$. (Shows how well execution matched plan.)

- **Digital Data Capture Rate (%):** $(\text{Events logged digitally} \div \text{Total events expected}) \times 100$. (Ensures the digital system is capturing reality.)
- **Safety Incident Rate (TRIFR):** $(\text{Total recordable injuries} \times 1,000,000) \div \text{Total man-hours worked}$. (Standard safety metric.)

These KPIs cover production, cost and environmental performance. Not every mine will use every metric, but a balanced set should be tracked. For example, cost per tonne and fuel per tonne combine a financial view with a carbon view. Improvement targets should be set for each KPI (e.g., “reduce fuel intensity by 5% in 12 months”) and progress reviewed frequently. Over time, adding or refining KPIs (as suggested by experience and data availability) ensures the mine focuses on what matters most.

Case Examples and Applied Tactics

Concrete examples help crystallize the concepts. The following cases illustrate how specific digital tactics improve outcomes:

- **Digital SIC Increases Throughput:** A coal mining site deployed short-interval control with mobile tracking. Previously, night shifts often underperformed due to communication lags. With digital SIC, each shift’s plan and results were logged in real time. Within weeks, the night shift’s productivity matched the day’s, reducing idle trucks by 25%. Overall coal deliveries rose by 12% with the same equipment, illustrating how digital discipline translates directly to output gains.
- **Precision Drilling for Efficiency:** A copper mine equipped with its drills with high-precision GPS guidance. Operators followed a

GPS overlay on a tablet to place each hole. The result was more uniform blast fragmentation. Loaders reported 10% faster loading rates, and primary crusher throughput increased. Indirectly, the mine used about 8% less diesel per tonne because loading and crushing was smoother. This case shows how digital drill guidance can cut both time and fuel.

- **Predictive Maintenance Avoids Failures:** A nickel mine added vibration monitors to key haul trucks. The system predicted a front differential bearing failure a week before the next scheduled service. Instead of an unexpected breakdown on a peak day, the truck was taken in for overhaul during a planned downtime. This kept the truck operating for its entire planned life and prevented an estimated 1000-liter fuel waste (from being idled) as well as millions of dollars in replacement cost.
- **Fuel Analytics and Ramp Redesign:** At a gold mine, analysis of fuel data showed that loaded trucks on one steep ramp burned 20% more diesel than on other routes. The mine rebuilt that ramp with a gentler slope and the fuel efficiency on that route improved by 10%. This demonstrates how data can pinpoint issues and how operational adjustments can yield quick environmental benefits.

These examples emphasize a pattern: use data to identify inefficiency or risk, then act on it. They also highlight cross-functional collaboration: maintenance, operations and engineering all work together. When similar tactics are applied broadly – for fleet, drilling, water, maintenance – the cumulative effect is transformational.

Conclusion – The Business Case for Digitizing for Green Outcomes

At first glance, digitizing mining sounds like a technical expense. However, the business case for linking digitization to green outcomes is strong. Here's why the investment pays off:

- **Cost Savings:** The efficiency improvements driven by digital tools quickly reduces costs. For example, even 1% fuel saving on a large fleet is substantial. Maintenance savings and higher throughput directly increase profit margins.
- **Higher Production:** Digital discipline and automation often boost output without additional capital. More tonnes produced per day means more revenue using the same equipment. This amplified production can dwarf the marginal cost of new technology.
- **Regulatory Compliance:** Many jurisdictions now require detailed environmental reporting. Having digital records ensures compliance and avoids fines. Plus, companies may receive incentives (tax credits, grants) for green upgrades, offsetting costs.
- **Risk Mitigation:** Accidents, spills or regulatory breaches are extremely costly. Digital systems act like insurance by catching issues early. Averted disaster can save the company millions in liabilities and preserve its social license.
- **Investor and Market Access:** Mining is a global business. Buyers and financiers prefer sustainably sourced minerals. Digitally verified ESG data can open doors to premium markets or green financing. In tough markets, a strong ESG profile can protect stock value.

- **Employee and Community Relations:** A safer, more modern workplace helps retain skilled staff. Good environmental practices support community relations, reducing the risk of local opposition or labor unrest that can halt production.
- **Futureproofing:** Global trends point to stricter environmental standards ahead. Mines that are digitized now will be ready for tomorrow's requirements. Lagging risks being uncompetitive or obsolete.



About GroundHog

Groundhog is a unified solution for digitizing and automating mining operations. The platform seamlessly integrates various workflows, spanning drill, blast, load, and haul processes, to enhance productivity, efficiency, safety and minimize variance to plan. At GroundHog, we are at the forefront of High Precision and Standard Precision GPS technology, empowering mining operations worldwide. Our state-of-the-art solutions are designed to deliver exceptional accuracy, reliability, and performance. With our GPS based Fleet Management System, you can unlock the full potential of your mining operations, optimize productivity, drive sustainable growth and enable Green Mining. Ready to experience the power of GroundHog Mine Digitization and Automation Solution with various functionalities?

Contact us today to schedule a personalized demo of our innovative solutions. Our team of experts will guide you through the features, benefits, and capabilities of our technology, tailored specifically to your mining needs. Don't miss out on this opportunity to revolutionize your mining operations with GroundHog Mine Digitization and Automation Solution with various functionalities. Get in touch with us now!

Published By:



groundhogApps.com

Published By:

GroundHog